

Evaluated Nuclear Structure Data File (ENSDF) Spin-Parity Instructions for GitHub Copilot

L-Transfer Field Positioning

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- L always starts from column 56 (EXACT rule for L transferred angular momentum fields)
- $L=1 \rightarrow 1$ at column 56
- $L=1+2 \rightarrow 1$ at column 56, $+2$ at columns 57-58
- $L=1,2 \rightarrow 1$ at column 56, $,2$ at columns 57-58
- ONLY the first L-value must be at column 56; subsequent values follow sequentially

L-Transfer from $0+$ for J- π Assignment Rules

Physics mapping (single-nucleon transfer from a $0+$ even-even target):

- $j = L \pm 1/2$
- $\pi = (-1)^L$

L-Transfer values and corresponding J- π assignments:

- $L=0 \rightarrow J-\pi: 1/2+$
- $L=1 \rightarrow J-\pi: 1/2-, 3/2-$
- $L=2 \rightarrow J-\pi: 3/2+, 5/2+$
- $L=3 \rightarrow J-\pi: 5/2-, 7/2-$
- $L=4 \rightarrow J-\pi: 7/2+, 9/2+$

J- π Assignment Notation

Fundamental Rule

CRITICAL: J = spin; π = parity

- **WITHOUT parentheses:** Firm, well-established assignments (e.g., $3/2+$, $7/2-$)
- **WITH parentheses:** Less certain, tentative assignments (e.g., $(3/2+)$, $(7/2-)$)
- **Parentheses indicate uncertainty in the assignment confidence, not the measurement precision**
- **NEVER change parentheses notation without experimental justification**

Complete J- π Notation Patterns

Basic Single Assignments

- $1/2-$ = firm spin-parity, no other possibilities
- $(9/2+)$ = tentative spin-parity, other spin-parity possibilities exist
- $7/2(+)$ = firm spin with tentative parity, other spin impossible; other parity possible
- $(5/2)-$ = tentative spin with firm negative parity, other spin possible; positive parity impossible

- $(11/2)$ = tentative spin, parity unknown or undetermined
- $+$ = only positive parity determined, spin unknown
- $-$ = only negative parity determined, spin unknown
- $(+)$ = tentative positive parity, spin unknown
- $(-)$ = tentative negative parity, spin unknown

Multiple Possible Assignments

- $1/2-, 3/2-$ = multiple spins with firm parity (both certain, comma-separated), no other spin-parity possibilities
- $(5/2+, 7/2+)$ = multiple tentative possibilities, other spin-parity possibilities exist
- $(1/2, 3/2)+$ = multiple tentative spins with firm positive parity, other spins possible; negative parity impossible
- $(7/2, 9/2)-$ = multiple tentative spins with firm negative parity, other spins possible; positive parity impossible
- $(1/2, 3/2, 5/2)-$ = multiple tentative spins with firm negative parity, other spins possible; positive parity impossible
- $(3/2, 5/2, 7/2+)$ = mixed notation: first two spins with both parities possible, last spin with firm positive parity

Range Assignments

- $(1/2+: 7/2+)$ = range of tentative spin-parity assignments from $1/2+$ to $7/2+$
- $(1/2: 9/2)-$ = range of tentative spins from $1/2$ to $9/2$ with firm negative parity
- $1/2+: 5/2+$ = range of firm spin-parity assignments from $1/2+$ to $5/2+$
- $(3/2: 11/2)$ = range of tentative spins from $3/2$ to $11/2$, parity undetermined

Mixed Confidence Patterns

- $3/2-, (5/2-)$ = first assignment firm, second tentative
- $(7/2)+, 9/2+$ = first assignment tentative, second firm
- $1/2(+), 3/2-$ = first has tentative parity, second fully firm
- $(5/2)+, (7/2)-$ = multiple assignments with different confidence levels

Special Cases

- $1/2+, 3/2-$ = multiple firm assignments with different parities
- $(5/2+, 7/2-)$ = multiple tentative assignments with different parities
- $3/2, 5/2, 7/2$ = multiple possible spins, parity undetermined
- $(1/2, 3/2, 5/2)$ = multiple tentative spins, parity undetermined

Critical Formatting Rules

- **Comma separation** for multiple possibilities within same confidence level
- **Parentheses apply to the entire group** when wrapping multiple values
- **Do not include explicit parity on an item inside a grouped tentative list**; if a subset has a different parity or confidence, split it out of the group

- **Mixed notation is allowed** by splitting: e.g., $1/2, 3/2, 5/2+$ (first two tentative, parity unspecified; last firm positive)
- **No spaces** around commas in J- π field
- **Exact reproduction required** - never modify parentheses placement without experimental justification

Critical Parentheses Matching Rule

Spin-parity with/without parentheses are considered to be different confidence levels. When creating J\$ comments or adding values to J fields from reference data sources, ensure parentheses are preserved exactly as written in the source:

- **Source shows $3/2$** → Comment: J\$ $3/2$ from [reference] (NO parentheses)
- **Source shows $(3/2)$** → Comment: J\$($3/2$) from [reference] (single parentheses preserved)
- **NEVER use double parentheses:** J\$(($3/2$)) is FORBIDDEN
- **Examples:** $(1/2+)$, $1/2(+)$, $1/2+$ represent different assignment confidence levels and the placement of parentheses must be matched accurately and precisely!